

Worksheet - Lecture 1

Introduction to Linear Algebra Part One

1. Use the following matrices or vectors to answer the following questions:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 8 \\ 3 & 0 & -2 \\ 4 & 1 & -3 \end{pmatrix} \quad \mathbf{M} = \begin{pmatrix} 1 & 8 & -2 & 5 \\ 2 & 8 & 1 & 7 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\mathbf{X} = \begin{pmatrix} 780 & 95000 \\ 600 & 60000 \\ 550 & 65000 \\ 400 & 35000 \\ 450 & 40000 \\ 750 & 80000 \end{pmatrix} \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1.3 \\ 0.8 \\ 2 \\ 2.5 \\ 0.8 \\ 0.9 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 6 \\ 3 \\ -1 \\ 2 \end{pmatrix} \quad \mathbf{u} = (6 \ 4 \ 8 \ 1)$$

a. Write the appropriate size/dimensions next to each matrix:

$$\mathbf{A} \ 3 \times 3$$

$$\mathbf{X} \ 6 \times 2$$

$$\mathbf{u} \ 1 \times 4$$

$$\mathbf{M} \ 2 \times 4$$

$$\mathbf{t} \ 7 \times 1$$

$$\mathbf{D} \ 3 \times 3$$

$$\mathbf{v} \ 4 \times 1$$

b. Which of these matrices are square? Which are rectangular?

A, D

all others

c. Give the following quantities:

$$A_{12} = 3$$

$$\mathbf{M}_{2*} = (2 \ 8 \ 1 \ 7)$$

$$v_3 = -1$$

$$M_{21} = 8$$

$$X_{42} = 35000$$

$$\mathbf{D}_{*3} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$$t_5 = 2.5$$

c. What are the diagonal elements of $\mathbf{A}$?

$$A_{11} = 1$$

$$A_{22} = 0$$

$$A_{33} = -3$$

2. For the following quantities, use what you know about notation to tell if they are matrices, vectors, or scalars:

$$\mathbf{H} \ \text{matrix}$$

$$\mathbf{v}_2 \ \text{vector}$$

$$\lambda \ \text{scalar}$$

$$\mathbf{W} \ \text{matrix}$$

$$v_2 \ \text{scalar}$$

$$A_{ij} \ \text{scalar}$$

$$n \ \text{scalar}$$

$$\mathbf{M}_{*2} \ \text{vector}$$

$$\mathbf{r} \ \text{vector}$$

Worksheet - Lecture 2

Introduction to Linear Algebra Part Two

1. Use the following matrices or vectors to answer the following questions:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 8 \\ 3 & 0 & -2 \\ 8 & -2 & -3 \end{pmatrix} \quad \mathbf{M} = \begin{pmatrix} 1 & 8 & -2 & 5 \\ 2 & 8 & 1 & 7 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} 6 \\ 3 \\ -1 \\ 2 \end{pmatrix} \quad \mathbf{u} = (6 \ 4 \ 8 \ 1)$$

a. Write the following matrices or vectors:

$$\mathbf{M}^T = \begin{pmatrix} 1 & 2 \\ 8 & 8 \\ -2 & 1 \\ 5 & 7 \end{pmatrix}$$

$$\mathbf{v}^T = (6 \ 3 \ -1 \ 2)$$

$$\mathbf{A}^T = \begin{pmatrix} 1 & 3 & 8 \\ 3 & 0 & -2 \\ 8 & -2 & -3 \end{pmatrix}$$

$$\mathbf{u}^T = \begin{pmatrix} 6 \\ 4 \\ 8 \\ 1 \end{pmatrix}$$

$$(\mathbf{M}^T)^T = \begin{pmatrix} 1 & 8 & -2 & 5 \\ 2 & 8 & 1 & 7 \end{pmatrix}$$

b. Which of these matrices is symmetric? **A**

c. Can a rectangular matrix be symmetric? **no. A^T cannot equal A unless these matrices have the same size!**

d. What is $\text{Tr}(\mathbf{A})$?

$$\text{Tr}(\mathbf{A}) = 1 + 0 + (-3) = -2$$

e. If possible, compute $\text{Tr}(\mathbf{M})$. If not possible, explain why.

not possible. Trace is undefined for square matrices

2. Write out the following matrices or vectors:

$$\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{e}_3 \in \mathbb{R}^5 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{diag}\{\sigma_1, \sigma_2, \sigma_3\} = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix}$$

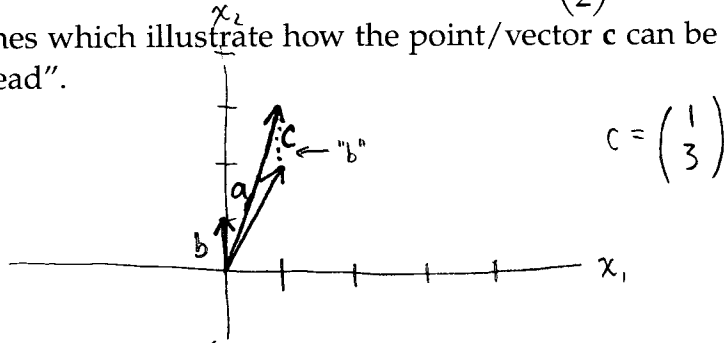
3. If a matrix $\mathbf{M}$ is upper triangular then $\mathbf{M}^T$ is lower triangular

4. If $\mathbf{S}$ is a diagonal matrix, then $S_{12} = \underline{0}$

Worksheet - Lectures 3 & 4

Matrix Arithmetic

1. On a coordinate plane, draw the vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and then draw $\mathbf{c} = \mathbf{a} + \mathbf{b}$. Make dotted lines which illustrate how the point/vector $\mathbf{c}$ can be reached by connecting the vectors $\mathbf{a}$ and $\mathbf{b}$ "tail-to-head".



2. Use the following vectors to answer the questions:

$$\mathbf{v} = \begin{pmatrix} 6 \\ -1 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} \quad \mathbf{z} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{oops!}$$

- a. Compute the following linear combinations, if possible:

$$2\mathbf{u} + 3\mathbf{v} = 2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 6 \\ -1 \end{pmatrix} = \begin{pmatrix} 14 \\ -1 \end{pmatrix} \quad \mathbf{u} + \mathbf{z} = \text{not possible}$$

$$\mathbf{x} - 2\mathbf{y} + \mathbf{z} = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} - 2 \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \\ 8 \end{pmatrix}$$

$$-2\mathbf{u} - \mathbf{v} + \mathbf{z} = \text{not possible}$$

- b. Compute the following inner products, if possible:

$$\mathbf{u}^T \mathbf{v} = (-2 \ 1) \begin{pmatrix} 6 \\ -1 \end{pmatrix} = -13 \quad \mathbf{x}^T \mathbf{u} = \text{not possible} \quad \mathbf{v}^T \mathbf{x} = \text{not poss}$$

$$\mathbf{x}^T \mathbf{x} = (4 \ 2 \ 1) \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = 21 \quad \mathbf{x}^T \mathbf{z} = (4 \ 2 \ 1) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 7 \quad \mathbf{z}^T \mathbf{v} = \text{not poss.}$$

$$\mathbf{z}^T \mathbf{y} = (1 \ 1 \ 1) \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} = -6 \quad \mathbf{y}^T \mathbf{z} = (-1 \ -2 \ -3) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -6$$

Worksheet - Lecture 4

Matrix Arithmetic Part Two

1. Use the following matrices to answer the questions:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 8 \\ 3 & 0 & -2 \\ 8 & -2 & -3 \end{pmatrix} \quad \mathbf{M} = \begin{pmatrix} 1 & 8 & -2 & 5 \\ 2 & 8 & 1 & 7 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\mathbf{H} = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} \quad \mathbf{W} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \end{pmatrix}$$

a. Circle the matrix products that are possible and specify their resulting dimensions:

$$\mathbf{AM} \quad 3 \times 3 \quad 2 \times 4$$

$$\mathbf{W}^T \mathbf{D} \quad 4 \times 3 \quad 3 \times 3$$

$$\mathbf{M}^T \mathbf{H}^T \quad 4 \times 2 \quad 2 \times 2$$

$$\mathbf{AW} \quad 3 \times 3 \quad 3 \times 4$$

$$\mathbf{HM} \quad 2 \times 2 \quad 2 \times 4$$

$$\mathbf{WD} \quad 3 \times 4 \quad 3 \times 3$$

$$\mathbf{MH} \quad 2 \times 4 \quad 2 \times 2$$

$$\mathbf{DW} \quad 3 \times 3 \quad 3 \times 4$$

• Compute the following matrix products:

HM and AD

$$\mathbf{HM} = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 8 & -2 & 5 \\ 2 & 8 & 1 & 7 \end{pmatrix} = \begin{pmatrix} 0 & 8 & -5 & 3 \\ 7 & 32 & 1 & 26 \end{pmatrix}$$

$$\mathbf{AD} = \begin{pmatrix} 1 & 3 & 8 \\ 3 & 0 & -2 \\ 8 & -2 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 15 & 24 \\ 3 & 0 & -6 \\ 8 & -10 & -9 \end{pmatrix}$$

• From the previous computation, **AD**, do you notice anything interesting about multiplying a matrix by a diagonal matrix on the right? Can you generalize what happens in words?

The columns of the matrix **A** will always be scaled by the corresponding diagonal element of **D**.

Different Views of Matrix Multiplication

2. Consider the matrix product $\mathbf{AB}$ where

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$$

Let $\mathbf{C} = \mathbf{AB}$.

- Compute the matrix product $\mathbf{C}$.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 11 \\ 10 & 27 \end{pmatrix}$$

- Compute the matrix-vector product $\mathbf{AB}_{*1}$ and show that this is the first column of $\mathbf{C}$. (Likewise, $\mathbf{AB}_{*2}$ is the second column of $\mathbf{C}$.) (Matrix multiplication can be viewed as a collection of matrix-vector products.)

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \end{pmatrix}$$

Likewise $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 11 \\ 27 \end{pmatrix}$

- Compute the two outer products using columns of $\mathbf{A}$ and rows of $\mathbf{B}$ and show that

$$\mathbf{A}_{*1}\mathbf{B}_{1*} + \mathbf{A}_{*2}\mathbf{B}_{2*} = \mathbf{C}$$

(Matrix multiplication can be viewed as the sum of outer products.)

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 & 5 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 6 & 15 \end{pmatrix} + \begin{pmatrix} 2 & 6 \\ 4 & 12 \end{pmatrix} = \begin{pmatrix} 4 & 11 \\ 10 & 27 \end{pmatrix}$$

- Since $\mathbf{AB}_{*1}$ is the first column of $\mathbf{C}$, show how $\mathbf{C}_{*1}$ can be written as a linear combination of columns of $\mathbf{A}$. (Matrix multiplication can be viewed as a collection of linear combinations of columns of the first matrix.)

$$\mathbf{C}_{*1} = B_{11} \mathbf{A}_{*1} + B_{21} \mathbf{A}_{*2}$$

$$\begin{pmatrix} 4 \\ 10 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

- Finally, note that $\mathbf{A}_{1*}\mathbf{B}$ will give the first row of $\mathbf{C}$. (This amounts to a linear combination of rows - can you see that?)

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 11 \end{pmatrix}$$

$$\Rightarrow 1 \cdot \begin{pmatrix} 2 & 5 \end{pmatrix} + 2 \cdot \begin{pmatrix} 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 11 \end{pmatrix} \quad \text{Linear Comb. of rows}$$

$$A_{11} \mathbf{B}_{1*} + A_{12} \mathbf{B}_{2*} = \mathbf{C}_{1*}$$

Worksheet - Lecture 5

Applications of Matrix Arithmetic

1. A florist offers three sizes of flower arrangements (small, medium, large) containing three types of flowers (roses, daisies, and chrysanthemums). The number of each type of flower in each size arrangement is given in the table below, along with the selling price of each arrangement and the cost of each individual flower.

	Roses	Daisies	Chrys.	Price
Small	1	3	3	\$10
Medium	2	4	6	\$15
Large	4	8	6	\$20
Cost	\$0.50	\$0.25	\$0.10	

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 3 \\ 2 & 4 & 6 \\ 4 & 8 & 6 \end{pmatrix} \quad \mathbf{p} = \begin{pmatrix} 10 \\ 15 \\ 20 \end{pmatrix} \quad \mathbf{c} = \begin{pmatrix} 0.50 \\ 0.25 \\ 0.10 \end{pmatrix}.$$

- a. Determine the matrix-vector product that produces a vector, $\mathbf{y}$ which gives the total cost of creating each size arrangement (small, medium, and large).

$$\mathbf{y} = \mathbf{A} \mathbf{c}$$

- b. Suppose that an order came in for 2 small arrangements and 2 large arrangements. Let

$$\mathbf{v} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

Using matrix arithmetic (and writing out the formula) determine both the price of this order and the total profit to the florist.

$$\text{price} = \mathbf{v}^T \mathbf{p} \quad (\text{or equivalently } \mathbf{p}^T \mathbf{v})$$

$$\text{cost} = \mathbf{v}^T \mathbf{y} = \mathbf{v}^T \mathbf{A} \mathbf{c}$$

$$\boxed{\text{profit} = \mathbf{v}^T \mathbf{p} - \mathbf{v}^T \mathbf{A} \mathbf{c}}$$

2. Write the following system of equations as a matrix-vector product $\mathbf{Ax} = \mathbf{b}$:

$$\begin{cases} 2x_2 + 3x_3 = 8 \\ 2x_1 + 3x_2 + 1x_3 = 5 \\ x_1 - x_2 - 2x_3 = -5 \end{cases}$$

$$\underbrace{\begin{pmatrix} 0 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & -1 & -2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 8 \\ 5 \\ -5 \end{pmatrix}}_{\mathbf{b}}$$

3. A model is being developed to predict a student's SAT score based upon some numeric attributes. The data being used for this model is provided below:

Observation	PSAT score	Mother's SAT score	SAT Score
1	1600	1700	1750
2	1800	1250	1750
3	1750	1300	1600
4	1200	1800	1450
5	1350	1950	1500

If our regression model is

$$SAT_score = \beta_0 + \beta_1 * PSAT_score + \beta_2 * Mothers_SAT_Score + \epsilon$$

So how we'd set up the underlying matrix equation for regression analysis,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$$

by defining the matrices/vectors $\mathbf{X}$, $\boldsymbol{\beta}$, and $\mathbf{y}$.

$$\mathbf{X} = \begin{pmatrix} 1 & 1600 & 1700 \\ 1 & 1800 & 1250 \\ 1 & 1750 & 1300 \\ 1 & 1200 & 1800 \\ 1 & 1350 & 1950 \end{pmatrix}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

$$\mathbf{y} = \begin{pmatrix} 1750 \\ 1750 \\ 1600 \\ 1450 \\ 1500 \end{pmatrix}$$

Worksheet - Lecture 6

Gaussian Elimination

1. Using Gaussian Elimination on the augmented matrices, reduce each system of equations to a triangular form and solve using back-substitution.

a.

$$\begin{cases} x_1 + 2x_2 = 3 \\ -x_1 + x_2 = 0 \end{cases}$$

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ -1 & 1 & 0 \end{array} \right) \xrightarrow{R_2' = R_2 + R_1} \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 3 & 3 \end{array} \right) \rightarrow \begin{aligned} x_1 + 2x_2 &= 3 \\ 3x_2 &= 3 \\ \Rightarrow x_2 &= 1 \\ x_1 + 2(1) &= 3 \\ \Rightarrow x_1 &= 1 \end{aligned}$$

b.

$$\begin{cases} x_1 + x_2 + 2x_3 = 7 \\ x_1 + x_3 = 4 \\ -2x_1 - 2x_2 = -6 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 1 & 0 & 1 & 4 \\ -2 & -2 & 0 & -6 \end{array} \right) \xrightarrow{\substack{R_2' = R_2 - R_1 \\ R_3' = R_3 + 2R_1}} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 4 & 8 \end{array} \right) \rightarrow \begin{aligned} x_1 + x_2 + 2x_3 &= 7 \\ -x_2 - x_3 &= -3 \\ 4x_3 &= 8 \\ \Rightarrow x_3 &= 2 \\ -x_2 - (2) &= -3 \\ \Rightarrow x_2 &= 1 \\ x_1 + (1) + 2(2) &= 7 \Rightarrow x_1 = 2 \end{aligned}$$

c.

$$\begin{cases} 2x_1 - x_2 + x_3 = 1 \\ -x_1 + 2x_2 + 3x_3 = 6 \\ x_2 + 4x_3 = 6 \end{cases}$$

$$\left(\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ -1 & 2 & 3 & 6 \\ 0 & 1 & 4 & 6 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} -1 & 2 & 3 & 6 \\ 2 & -1 & 1 & 1 \\ 0 & 1 & 4 & 6 \end{array} \right)$$

$$\xrightarrow{R_2' = R_2 + 2R_1} \left(\begin{array}{ccc|c} -1 & 2 & 3 & 6 \\ 0 & 3 & 7 & 13 \\ 0 & 1 & 4 & 6 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} -1 & 2 & 3 & 6 \\ 0 & 1 & 4 & 6 \\ 0 & 3 & 7 & 13 \end{array} \right)$$

$$\xrightarrow{R_3' = R_3 - 3R_2} \left(\begin{array}{ccc|c} -1 & 2 & 3 & 6 \\ 0 & 1 & 4 & 6 \\ 0 & -5 & -5 & -5 \end{array} \right) \Rightarrow \boxed{x_3 = 1} \Rightarrow \boxed{x_2 = 2} \Rightarrow \boxed{x_1 = 1}$$

Worksheet - Lecture 7

Gauss-Jordan Elimination

1. Using Gauss-Jordan Elimination on the augmented matrices, reduce each system of equations from the previous worksheet (listed again below) to reduced row-echelon form and give the solution as a vector.

a. Starting from previous worksheet solution

$$\begin{cases} x_1 + 2x_2 = 3 \\ -x_1 + x_2 = 0 \end{cases}$$

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 3 & 3 \end{array} \right) \xrightarrow{R_2' = \frac{1}{3} R_2} \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 1 \end{array} \right) \xrightarrow{R_1' = R_1 - 2R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right)$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

b.

$$\begin{cases} x_1 + x_2 + 2x_3 = 7 \\ x_1 + x_3 = 4 \\ -2x_1 - 2x_2 = -6 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 4 & 8 \end{array} \right) \xrightarrow{\substack{R_3' = \frac{1}{4} R_3 \\ R_2' = -1 R_2}} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

$$\xrightarrow{\substack{R_2' = R_2 - R_3 \\ R_1' = R_1 - 2R_3}} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right) \xrightarrow{R_1' = R_1 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

c.

$$\begin{cases} 2x_1 - x_2 + x_3 = 1 \\ -x_1 + 2x_2 + 3x_3 = 6 \\ x_2 + 4x_3 = 6 \end{cases}$$

$$\left(\begin{array}{ccc|c} -1 & 2 & 3 & 6 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & -5 & -5 \end{array} \right) \xrightarrow{\substack{R_1' = -1 R_1 \\ R_3' = -\frac{1}{5} R_3}} \left(\begin{array}{ccc|c} 1 & -2 & -3 & -6 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

$$\xrightarrow{\substack{R_2' = R_2 - 4R_3 \\ R_1' = R_1 + 3R_3}} \left(\begin{array}{ccc|c} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow{R_1' = R_1 + 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Worksheet - Lecture 8

Characterizing Solution Sets

1. For each of the following systems, determine whether they have a unique solution, no solution, or infinitely many solutions. If a system has a unique solution, provide that solution. If a system has infinitely many solutions, characterize the solution set as was done in the lecture.

a.
$$\begin{cases} x + 2y + z = 2 \\ 2x + 4y = 2 \\ 3x + 6y + z = 4 \end{cases}$$

$$\begin{pmatrix} 1 & 2 & 1 & | & 2 \\ 2 & 4 & 0 & | & 2 \\ 3 & 6 & 1 & | & 4 \end{pmatrix} \xrightarrow{\substack{R_1' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1}} \begin{pmatrix} 1 & 2 & 1 & | & 2 \\ 0 & 0 & -2 & | & -2 \\ 0 & 0 & -2 & | & -2 \end{pmatrix}$$

$$\xrightarrow{\substack{R_3' = R_3 - R_2 \\ R_2' = -\frac{1}{2}R_2}} \begin{pmatrix} 1 & 2 & 1 & | & 2 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \xrightarrow{R_1' = R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & | & 1 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$x_1 + 2x_2 = 1$
 $x_3 = 1$
 $x_1 = 1 - 2x_2$
 $x_2 = \text{free}$
 $x_3 = 1$
 let $x_2 = s$
 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1-2s \\ s \\ 1 \end{pmatrix}$

b.
$$\begin{cases} 2x_1 + 2x_2 + 6x_3 = 4 \\ 2x_1 + x_2 + 7x_3 = 6 \\ -2x_1 - 6x_2 - 7x_3 = -1 \end{cases}$$

$$\begin{pmatrix} 2 & 2 & 6 & | & 4 \\ 2 & 1 & 7 & | & 6 \\ -2 & -6 & -7 & | & -1 \end{pmatrix}$$

unique solution: $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

c.
$$\begin{cases} -h + l + w = 1 \\ -h - l + w = 2 \\ h - l + w = 3 \\ h + l + w = 4 \end{cases}$$

$$\begin{pmatrix} -1 & 1 & 1 & | & 1 \\ -1 & -1 & 1 & | & 2 \\ 1 & -1 & 1 & | & 3 \\ 1 & 1 & 1 & | & 4 \end{pmatrix} \xrightarrow{\substack{R_2' = R_2 - R_1 \\ R_3' = R_3 + R_1 \\ R_4' = R_4 + R_1}} \begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 2 & 2 & | & 5 \end{pmatrix} \xrightarrow{R_4' = R_4 + R_2} \begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 0 & 2 & | & 6 \end{pmatrix}$$

$$\xrightarrow{R_4' = R_4 - R_3} \begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 0 & 0 & | & 2 \end{pmatrix}$$

INCONSISTENT

d.
$$\begin{cases} -h + l + w = 1 \\ -h - l + w = 2 \\ h - l + w = 3 \\ h + l + w = 2 \end{cases}$$

Same steps as above, only last row changes.
 reach $\begin{pmatrix} -1 & 1 & 1 & | & 1 \\ 0 & -2 & 0 & | & 1 \\ 0 & 0 & 2 & | & 4 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 2 \end{pmatrix}$ unique solution

e.
$$\begin{cases} x_1 + 2x_2 + 2x_3 + 3x_4 = 0 \\ 2x_1 + 4x_2 + x_3 + 3x_4 = 0 \\ 3x_1 + 6x_2 + x_3 + 4x_4 = 0 \end{cases}$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 2 & 4 & 1 & 3 & | & 0 \\ 3 & 6 & 1 & 4 & | & 0 \end{pmatrix} \xrightarrow{\substack{R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1}} \begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 0 & 0 & -3 & -3 & | & 0 \\ 0 & 0 & -5 & -5 & | & 0 \end{pmatrix}$$

$$\xrightarrow{\substack{R_2' = -\frac{1}{3}R_2 \\ R_3' = -\frac{1}{5}R_3}} \begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \end{pmatrix} \xrightarrow{R_3' = R_3 - R_2} \begin{pmatrix} 1 & 2 & 2 & 3 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$$\xrightarrow{R_1' = R_1 - 2R_2} \begin{pmatrix} 1 & 2 & 0 & 1 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$x_1 = -x_2 - x_4$
 $x_2 = \text{free}$
 $x_3 = -x_4$
 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = s \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$

Worksheet - Lecture 9

Solving For A Matrix Inverse

1. Find the inverse of the following matrices, using the method of Gauss-Jordan Elimination to solve the matrix equation $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

a. $\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\boxed{\mathbf{A}^{-1} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -3/2 & 1 \\ 0 & 1 & -1 \end{pmatrix}}$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 & -3/2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 \end{array} \right)$$

b. $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1/2 \end{array} \right)$$

$$\Rightarrow \boxed{\mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1/2 \end{pmatrix}}$$

2. What is the inverse of a full rank diagonal matrix,

$$\mathbf{D} = \text{diag}\{d_{11}, d_{22}, \dots, d_{nn}\}$$

where each diagonal element, $d_{ii} \neq 0$?

$$\boxed{\mathbf{D}^{-1} = \text{diag} \left\{ \frac{1}{d_{11}}, \frac{1}{d_{22}}, \dots, \frac{1}{d_{nn}} \right\}}$$

Worksheet - Lecture 11

Least Squares

For the following problems you may use software or an online matrix inverse calculator to compute any matrix inverse required. One option can be found here:

<http://matrix.reshish.com/inverse.php>

- Find the least squares solution to following inconsistent system:

$$\underbrace{\begin{pmatrix} 2 & -2 \\ -2 & 2 \\ 5 & 3 \end{pmatrix}}_X \underbrace{\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}}_{\beta} = \underbrace{\begin{pmatrix} -1 \\ 7 \\ -26 \end{pmatrix}}_y$$

$$X^T X = \begin{pmatrix} 33 & 7 \\ 7 & 17 \end{pmatrix}$$

$$(X^T X)^{-1} = \begin{pmatrix} 0.0332 & -0.0137 \\ 0.0137 & 0.0645 \end{pmatrix}$$

$$X^T y = \begin{pmatrix} -146 \\ -62 \end{pmatrix}$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$= \begin{pmatrix} 0.0332 & -0.0137 \\ -0.0137 & 0.0645 \end{pmatrix} \begin{pmatrix} -146 \\ -62 \end{pmatrix}$$

$$= \begin{pmatrix} -4 \\ -2 \end{pmatrix}$$

- Use the method of least squares to find the regression equation associated with Problem 3 on Worksheet 5 (regarding SAT scores).

$$\hat{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 91.102 \\ 0.717 \\ 0.260 \end{pmatrix}$$

your answers may
be slightly different
depending on how
you round them

Worksheet - Lecture 12

Linear Independence

1. Determine whether or not the vectors

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

are linearly independent. Justify your answer using the definition of linear dependence/independence.

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \\ 3 & 1 & -1 & 0 \end{array} \right) \xrightarrow[\text{Jordan}]{\text{Gauss}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

Since the equation $\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \alpha_3 \mathbf{x}_3 = \mathbf{0}$ has only the trivial solution, $\alpha_1 = \alpha_2 = \alpha_3 = 0$, the vectors ARE linearly independent.

2. What is the rank of the matrix $\mathbf{M} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & -1 & 2 \\ 2 & 0 & 2 \end{pmatrix}$?

$$\left(\begin{array}{ccc} 2 & 1 & 0 \\ 0 & -1 & 2 \\ 2 & 0 & 2 \end{array} \right) \xrightarrow[\text{Jordan}]{\text{Gauss}} \left(\begin{array}{ccc} \textcircled{1} & 0 & 1 \\ 0 & \textcircled{1} & -2 \\ 0 & 0 & 0 \end{array} \right) \quad \text{Rank}(\mathbf{M}) = 2$$

- b. How many linearly independent rows does $\mathbf{M}$ have? How many linearly independent columns? 2, 2

- c. Is $\mathbf{M}$ full rank? no, $\text{rank} < 3$

3. (True/False) If $\mathbf{Ax} = \mathbf{b}$ has a solution then $\mathbf{b}$ can be written as a linear combination of the columns of $\mathbf{A}$.

True!!!